

Implementation Framework and Strategies for AI-augmented Open Educational Resources (OER): A Comprehensive Approach Applied to Secondary and Higher Education

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Abstract—This paper aims to propose an implementation framework for the adoption and management of Open Educational Resources (OER), focusing on their lifecycle, as well as the integration of AI for supporting educators in their classification of the created content, the creation of tutoring agents for the learning process and learners in deepening their learning experience and exploitation. We explore the incentives for educators, connections to educational programs, and propose a participatory design model for effective implementation. The application of this framework to the Graasp¹ learning experience platform and its associated open OER library is also discussed, along with future implementation strategies.

Keywords—digital education, open educational resources, OER, digital library, generative AI

I. INTRODUCTION

In this paper, we propose a comprehensive framework for managing Open Educational Resources (OER), addressing key aspects such as adoption by teachers, lifecycle management, and classification. Our goal is to outline practical strategies for implementing this framework in educational settings, focusing on AI tutoring agents particularly in secondary and higher education, and to demonstrate the feasibility of developing and deploying such an architecture.

A. Open Educational Resources

Open Educational Resources (OER) are teaching, learning, and research materials that are freely available in the public domain or under an open license permitting free use, adaptation, and distribution. OER encompasses a wide range of resources, including textbooks, course materials, lesson plans, multimedia content, and scholarly articles. Additionally, OER includes interactive elements such as animations, simulations, or serious games, as well as AI-supported chatbots and specifically tailored AI agents like

Socrates bots, which facilitate personalized learning experiences and enhance educational engagement.

B. AI-augmented OER

When an OER integrates interactive components and chatbots supported by Artificial Intelligence (AI), as well as embeds pedagogical scenarios and learning analytics, we refer to it as a rich OER (Fig. 1). In addition, OER can be stateless, i.e. just providing content to be consumed, or stateful, i.e. offering services enabling the individual or collaborative creation of learning outputs, as well as self-assessment and reflection through learning analytics. Stateful resources can only be used as part of learning platforms supporting user management and storage.

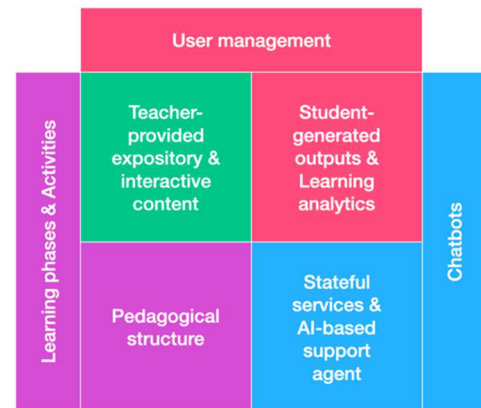


Fig. 1. Elements that can be part of a rich OER. Basic OER elements are represented in the green square. Elements bringing added pedagogical value are represented in magenta (pedagogical structure based on learning phases or activities). Elements relying on user management for analysing outputs and offering learning analytics are represented in red. Finally, elements relying on AI and enabling personalization and support through bots are represented in blue.

¹ <https://graasp.org>

AI-augmented OER might fundamentally expand and enhance the conceptual understanding of an educational resource, especially when relying on Generative (Gen) AI. First of all, it expands the possibility of on-the-fly created and situated feedback and interaction with the learner. The pedagogical added value ranges from potentially explaining all learning terms, concepts and theories adapted to the individual needs of the learner, to the generation of endless tailored examples and exercises including solutions about the topic. It might also create formative assessment sessions adapted to the individual needs of the learner. Specifically designed individual motivational hints and support can also be created on the fly and disseminated randomly throughout the learning journey. The learning design of a learning unit including Gen-AI-augmented educational resources can reach a better personalization of feedback and formative assessment within the existing learning material. A Gen-AI-augmented educational resource can adapt to beginner and expert learners' needs by creating tasks, examples and feedback loops of different complexity and difficulty.

C. State-of-the-art

Building on existing research [1][2][5][6], we review related work on OER adoption, metadata management, and the role of AI in educational resource classification. This section aims to establish the theoretical foundation for our proposed framework and highlight the current gaps in the literature.

According to UNESCO, OER are resources that are freely accessible and openly licensed, enabling educational institutions, educators, and learners to use, adapt, and share knowledge without cost or access barriers. This definition highlights the inclusive nature of OER, promoting educational equity and collaboration [1]. The integration of **interactive** elements in OER, such as animations, simulations, or serious games, is a way to fully embrace digitalization to bring added value compared to paper-based educational resources. Domains such as STEM education strongly benefit from interactive resources such as virtual labs [8]. The inclusion of AI-supported interactive elements to **enhance** OER is increasingly recognized in contemporary literature. Reference [2] discusses the integration of AI technologies in educational resources, emphasizing how chatbots and AI agents like Socrates bots can provide personalized tutoring, real-time feedback, and adaptive learning pathways. These advancements not only enrich the educational experience but also address diverse learning needs [2]. Reference [4] points out that the adaptability and interactivity of OER are key **benefits** that distinguish them from traditional educational materials. By incorporating AI-driven tools, OER can offer more engaging and effective learning experiences. Interactive elements such as simulations, virtual labs, and intelligent tutoring systems are cited as examples of how technology enhances the utility and impact of OER [4].

D. Rich and Stateful OER

The incorporation of AI-supported chatbots and tailored AI agents into OER signifies a shift towards more dynamic and responsive educational tools. For instance, a Socrates bot, designed to engage learners in Socratic questioning, can provide a unique, interactive learning experience that promotes critical thinking and deeper understanding. These AI agents can adapt to individual learners' pace and style, offering a customized educational journey. Furthermore, the use of learning analytics allows for the continuous

improvement of AI-generated content, ensuring it meets the evolving needs of learners.

Open educational resources are not limited to static content but include a broad spectrum of materials designed to support diverse educational needs. By embracing innovative technologies like AI-supported chatbots and specialized AI agents, OER continues to evolve, offering richer, more interactive, and personalized learning experiences. This integration ensures that OER remains at the forefront of educational innovation, fostering an inclusive and adaptive learning environment.

Examples of such educational AI-supported Bots (entities) have been published on the graasp.org open educational platform. Some entities provide additional examples tailored to the specific needs of the learners, some can be tailored to give adequate definitions of terms and build an individual glossary, some can act as tailored Socratic partners for learning (see link in Fig. 2).

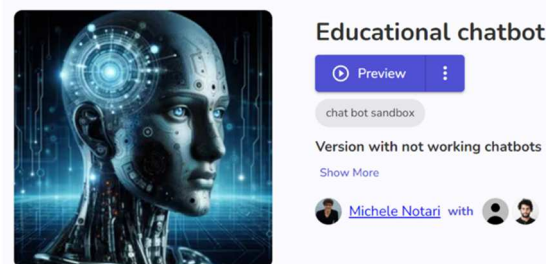


Fig. 2. Access to the different AI-bots using the [graasp.org](https://bit.ly/DmSSE) library (<https://bit.ly/DmSSE>).

E. OER Lifecycle

To really support education in the 21st century, it is not enough to just discuss OER as rich digital content part of the teaching and learning ecosystem. Despite the fact that the concept of OER is well known, there is still a huge scarcity in their availability. It is why it is important to consider the full OER lifecycle (Fig. 3), starting from i) encouraging, incentivizing, and supporting their creation, co-creation, and personalization; continuing with ii) promoting their usage in active blended teaching and learning scenarios; iii) promoting and facilitating their sharing and discovery in repositories and open libraries; and ending with iv) collecting and disseminating evidences of their adoption and impact.



Fig. 3. OER Lifecycle and its four core dimensions.

In the next Sections, we discuss the core dimensions of the OER lifecycle, from creation to dissemination and reuse. The focus is on the practical challenges teachers face in adopting OER and the support mechanisms required to facilitate this process.

II. INCENTIVES FOR TEACHERS

The use of OER by teachers can be significantly incentivized, especially when the resources can be searched by educational programs, topics, school levels, and resource types. Key incentives include flexibility, cost-effectiveness, professional development, efficiency, and alignment with educational equity.

A. Linking OER and Educational Programs

Linking OER to teachers' own educational programs can serve as a significant incentive. In Switzerland, there is a study plan per linguistic region, i.e. the *Plan d'études romand* (PER), the *Lehrplan 21*, and the *Piano di Studio*, respectively for the French, German, and Italian speaking region of the country. The *International Baccalaureate* is also largely used by international private schools in Switzerland. Using these study plans as references can enhance search and classification functionalities, making it easier for teachers to find relevant resources. This alignment ensures that teachers can efficiently find and use resources that match their specific teaching contexts and student needs, improving instructional effectiveness and student outcomes [1]. Strategies for integrating these programs with open libraries are essential for promoting flexibility, cost-effectiveness, and professional development among educators [2].

B. Participatory Design with Teachers

A participatory design approach involving teachers in developing OER frameworks is advocated based on previous research [3]. Engaging educators in the design process ensures that the resources created are relevant and practical, fostering a culture of continuous improvement and innovation in teaching. This approach also supports the creation of diverse and culturally relevant educational materials, enhancing inclusivity and representation in the classroom [5].

C. Incentives for Using OER

The incentives for teachers to use OER are multifaceted. The flexibility to tailor educational materials to specific teaching contexts allows for more effective and engaging instruction. The accessibility and adaptability of OER provide teachers with the flexibility to modify and adapt resources to address diverse learning styles and needs, fostering an inclusive classroom environment.

The cost-effectiveness of OER reduces financial barriers, particularly in underfunded educational settings, providing high-quality resources without additional costs [2]. Traditional educational resources, such as textbooks and proprietary software, can be prohibitively expensive, particularly for schools with limited budgets. By utilizing OER, teachers can provide high-quality educational materials without incurring additional costs, thereby reducing financial barriers for both educators and students [2].

Collaborative engagement with OER communities promotes professional development and peer-to-peer learning among teachers. Engaging with OER communities allows educators to share resources, teaching strategies, and best

practices, fostering a culture of continuous improvement and innovation in teaching [5].

The integration of advanced search functionalities in OER platforms, such as filtering by educational programs, topics, school levels, and resource types, streamlines the process of finding appropriate resources. This efficiency saves teachers valuable time and effort, allowing them to focus more on instructional design and student interaction [6]. Efficient access to resources is crucial for teachers who often work under significant time constraints.

AI-bots for individual tailored feedback or tutoring to the learner and support for the learning online journey expands and enhances the potential and usage opportunity of OER in multiple online and offline didactical settings.

The use of OER aligns with the principles of open education and equity. By adopting OER, teachers contribute to a more equitable educational landscape where high-quality resources are accessible to all students, regardless of their socio-economic background [1]. This democratization of knowledge supports broader educational goals and promotes lifelong learning.

III. CLASSIFICATION OF RESOURCES

As highlighted in the previous Sections, accessing OER is essential to promote and successfully implement digital education. However, the need to provide metadata or enable indexing to ease their sharing, search, and retrieval on open libraries and repositories is a huge hurdle for OER creators and consumers.

AI in general and Gen AI in particular is a great opportunity to tackle the issue of streamlining the process of organizing and accessing OER, by semi-automatically generating metadata or handling classification based on the information provided or extracted from the resources.

In this Section, after describing the classical classification using manually or semi-automatically captured metadata, we propose a way to create connections or a mapping to study plans using AI agents.

A. Typical OER Metadata

The classification of educational resources is based on the long and large experience of the brick-and-mortar libraries, as well as the experience in school and academic publishing. In this framework, references to resources usually include:

- Title;
- Author(s): Name, ID, affiliation, biography, ...
- Summary or description;
- Label(s): Keywords or tags;
- Language;
- Publisher: Company, institution, or organization, publication date, edition or version, and copyrights;
- Type: Book, map, movie, simulation, ...
- Category: Education, entertainment, novel, ...
- Target audience: General public, kids, adults, professional, academic, ...

In addition, for science and education, there are also:

- Framework: Conference, journal, collection, regional, national, or international context, ...;
- Discipline (subject domain) and sub-discipline (s);
- Level(s): Scholarly, educational level(s), or age range(s);

When referring to digital resources, there are in addition:

- Hosting media and size: CD, DVD (off-line), cloud (online), ...
- Typical duration of the activity associated with the use of the resource;
- Associated support material if not integrated in the resource;
- Preview or excerpt;
- Licensing rights, attributions;

Finally, with cloud libraries and repositories, the outcome of social filtering is also integrated:

- Popularity: Number of views, ratings, likes, ...
- Similar resources;
- Comments.

Previously, these metadata had, and still have in some cases, to be used for the classification of new resources by librarians and to enable users to find the resources. At the time of the first digital libraries, the need for teachers to provide such metadata before sharing resources was the main hurdle for promoting and developing a sharing culture. The word **metadata** became a word that it was better not to pronounce. Another word which also became scary for teachers was the term **ontology** used to represent and connect concepts, disciplines and subdisciplines. Fortunately, Google made all these challenges obsolete or invisible in 1998 and enabled users to share content without providing explicitly any metadata and ontology. Large digital libraries, like IEEE Xplore or Amazon, still provide some of these metadata explicitly to facilitate faceted and advanced searches, but resource providers do not have any more to worry about them, except mostly the 4 first ones.

As mentioned in Chapter II.A, teachers however really need to link the resources with their teaching framework and programs, especially with the teaching disciplines and levels for searching and implementing resources. However, these dimensions are strongly contextual and cannot easily be elicited by search engines and used for recommendation. It is where the potential of generative AI can deeply help, as detailed in the next chapter, and empower teachers like never before. Fig. 4 shows the metadata manually created in the framework of the Go-Lab European initiative on STEM education with online laboratories (labs). In this framework, the concept of big ideas of sciences was promoted, which was not possible to elicited automatically when they were introduced [8].

The general idea for associating a resource with teaching programs is to provide documents describing the programs of interest to a generative AI and to query the mapping of a given resource with its basic description to the relevant categories. The mapping is presented to the author of the resource who

can accept, reject or adapt the recommendation. It is what we call **classification recommendation**. Hence, it is a semi-automatic classification, making it easier for authors to contribute and for users to find relevant materials.

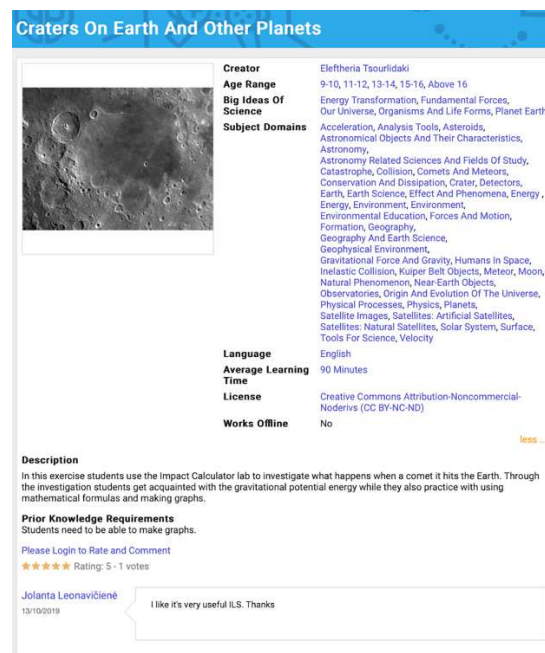


Fig. 4. Example of an Inquiry Learning Space (ILS) shared as an OER in the former Golabz.eu library for STEM education using virtual labs.

IV. CLASSIFICATION AND RECOMMENDATION

Before sketching in this Section how to support classification with Gen AI, it is important to highlight teachers' practices in terms of searching and sharing. The typical search behaviors are the following:

- Teachers get links to online resources from peers (word-of-mouth);
- Teachers use search engines and often end-up on commercial providers they cannot afford, either individually or through their institutions;
- Teachers go to their official school repositories or specialized libraries, which are often only showing a very small portion of all available educational resources, not leveraging on the collective intelligence and production of the world-wide teacher population. These repositories are also most of the time not accessible to learners and parents.
- Teachers are often not aware of newer technological and pedagogical developments of learning experiences.

There are no typical sharing behaviors, as it is still quite rare and challenging. The main limitations are:

- Teachers have limited means to share freely resources (only institutional learning management systems, platforms like YouTube, Google Drive, SlideShare, or wikis, ...);
- Sharing on school repositories are regulated and cumbersome, as only authorized people can publish and resources are most of the time manually evaluated before being published;

- Sharing on open libraries and repositories is closed, i.e. it can only be done by administrators, or is moderated, and require the teachers to provide a lot of information for validation and classification. Once published, the OER hosted on these infrastructures are usually pretty well referenced on search engines.

A. Classification Scheme

The general idea to streamline the sharing process is to let the teachers provide only the information they want. The rest of the information is automatically extracted from the content and the context of the resources themselves. For specific information like the link to a specific study plan, reference material can be submitted by the teachers and recommendation of the most relevant categories is proposed by the Gen AI service. Teachers can then select the categories they want to accept or personalize them. See the current dialog for publishing content in the open library associated with the grasp.org open access and open source learning experience platform in Fig. 5.

The mockup shows a 'Preview' section with a 'Saved' status. Below it, a description of the 'xVPC' resource is displayed. The 'Category' section includes 'vocation-training', 'informatics', and 'higher-education'. The 'Languages' section shows 'english'. The 'License' section displays 'Attribution-NonCommercial 4.0 International'. There are also buttons for 'Edit Tags', 'Value Proposition', 'Canvas', and 'Explainability'.

Fig. 5. Current dialog for publishing content in the open library associated with the grasp.org open access and open source learning experience platform (title, authors, language, description, disciplines from a fix arbitrary list, level, tags).

The mockup of the modified dialog to support the adjunction of semi-automatically generated categories is shown in Fig. 6. A reference document describing the relevant study plan can be provided, and then categories can be recommended by the embedded generative AI service. Then, the recommended categories can be accepted, rejected or modified (Fig. 7).

B. Validation Scheme

When enabling teachers to submit resources for publication in an open library without going through a human moderation process, it is also important to validate the submitted content for appropriateness. Large language models (LLMs) like *Gemini*, *OpenAI GPT* or open source models can be used to moderate content. Other specialized services like

Azure AI Content Safety, *OpenAI Moderation*, or *Google Cloud Text Moderation* offer more robust detection of harmful content and mature infrastructure. They come however with higher costs and data privacy concerns. LLMs, on the other hand, offer potential cost savings and data confidentiality through self-hosting, but face challenges with accuracy, prompt injection vulnerabilities, and complexity. One should mention that, in the case of OER, privacy is not a strong concern as the resources are anyway validated to be openly shared. What is important is to make users aware of the fact that the content is submitted to external services and to make sure these services respect the chosen creative commons license. Finally, it is worth mentioning that on open libraries and repositories with social features, ratings and likes can be used to make the best content visible, providing as such an indirect validation and hiding the low quality or inappropriate content if not detected otherwise.

The mockup shows a 'My OER' section with a 'Preview thumbnail' and 'Edit tag' button. Below it, a 'Description' field contains placeholder text. The 'Categories' section shows 'Resource type', 'Discipline', and 'Education level'. A 'Recommended categories' section includes a button to 'Provide a reference study plan (click or drag-and-drop): IB.pdf' and a 'Propose new' button.

Fig. 6. Mockup of the publication dialog integrating recommended categories based on Gen AI (green part). This view shows the first submission step, including the upload of a reference study plan.

The mockup shows a 'Categories' section with 'Resource type', 'Discipline', and 'Education level'. A 'Recommended categories' section includes a button to 'Provide a reference study plan (click or drag-and-drop): IB.pdf' and a 'Propose new' button. Below this, a section for 'New discipline', 'Sub discipline', and 'Other level' is shown, each with a button to 'Accept'.

Fig. 7. Mockup of the publication dialog integrating recommended categories based on Gen AI (bottom part). This view shows the proposed recommendation with options to reject or personalize.

Once accepted, the recommended categories are displayed together with the selection of predefined categories made by the users. Predefined categories are automatically translated if the users select another language for the user interface. Recommended categories proposed based on study plan documents are not translated, as the study plan might be only relevant in one language or one linguistic region.

V. CONCLUDING REMARKS

We conclude by summarizing the key findings and proposing future directions for research and implementation. The focus is on refining the framework based on feedback from educators and further integrating AI technologies for enhanced metadata management.

We propose to promote the development of an OER capsule to create tailored AI-augmented bots for specific didactical needs. The capsule is a sort of a reusable ‘template’ with the following characteristics. It must be conceived the way that an educator can easily tailor the AI-bot according to his target audience. The tailoring process must be simple, without any programming skills needed and include the possibility to integrate learning content and individual data of the learners gathered through learning analytics. Furthermore, the bot must insure that the data entered by the learner remain safe and not become part of the learning material for the Gen AI.

The classification process (AI-supported metadata collection) has to take place constantly during the creation of the learning sequence. The metadata should be proposed to the teacher creating learning content at the time of the publication of the material. The metadata related to the teaching program should be displayed to the teacher while the learning material is created in order to enhance the awareness of the educator toward specific needs to be addressed for the learners.

The opportunity to automatically classify the learning material is a step toward facilitating creation, sharing and finding of specific learning material. The open access to the learning material (OER) allows and reinforce to use, to adapt and to further share learning opportunities with other colleagues and the world.

The validation scheme proposed in section III-B will be further investigated in terms of classification accuracy and design appropriateness.

Having intelligent agents as tailored bots for specific didactical needs of a modern, interactive, individually adaptive learning experience opens the opportunity for deeper learning.

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